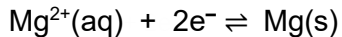
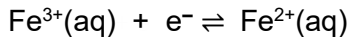


27 A voltaic cell is set up using the Mg^{2+}/Mg and $\text{Fe}^{3+}/\text{Fe}^{2+}$ half-cells.



$$E^{\ominus}_{\text{Mg}^{2+}/\text{Mg}} = -2.38 \text{ V}$$



$$E^{\ominus}_{\text{Fe}^{3+}/\text{Fe}^{2+}} = +0.77 \text{ V}$$

Under standard conditions, the cell e.m.f. would be 3.15 V. However, the voltmeter recorded a reading of 3.05 V.

What is the best explanation for this lower e.m.f.?

- 1 a higher concentration of Fe^{3+} was used
- 2 a higher concentration of Mg^{2+} was used
- 3 a smaller magnesium electrode was used

A 1 only

B 2 only

C 1 and 2 only

D 1, 2 and 3